

Fibonacci Series Generator

1. Prompt

Write a program to generate the **Fibonacci Series** up to n terms entered by the user.

The **Fibonacci Series** is a sequence where each number is the **sum of the previous two numbers**.

Example sequence:

0, 1, 1, 2, 3, 5, 8, 13, 21 ...

2. Steps / Algorithm

1. Start the program.
2. Ask the user to enter the number of terms n .
3. Initialize the first two numbers:
 - $a = 0$
 - $b = 1$
4. Print the first two numbers.
5. Use a loop from 2 to n .
6. Inside the loop:
 - Calculate the next term
 $c = a + b$
 - Print c

- Update values:

- `a = b`

- `b = c`

7. Repeat until the required number of terms is generated.
8. Stop the program.

3. Code (Python Program)

Fibonacci Series Generator

```
n = int(input("Enter the number of terms: "))
```

```
a = 0
```

```
b = 1
```

```
print("Fibonacci Series:")
```

```
for i in range(n):
```

```
    print(a, end=" ")
```

```
    c = a + b
```

```
    a = b
```

```
    b = c
```

4. Debugging Method

Debugging is the process of **finding and fixing errors in the program.**

Common Errors and Fixes

Error	Cause	Fix
ValueError	User enters text instead of number	Use integer input

Wrong output Logic mistake in formula

Check $c = a + b$

Infinite loop Incorrect loop condition

Ensure correct loop range

Example Debugging Version

```
n = int(input("Enter the number of terms: "))
```

```
a = 0
```

```
b = 1
```

```
print("Debug: Initial values -> a =", a, "b =", b)
```

```
for i in range(n):
```

```
    print("Term", i+1, ":", a)
```

```
    c = a + b
```

```
    print("Debug: Next value =", c)
```

```
    a = b
```

```
    b = c
```

This helps track how values change during execution.

5. Output

Example Output

Enter the number of terms: 8

Fibonacci Series:

0 1 1 2 3 5 8 13

Output Explanation

Term	Value
------	-------

1	0
---	---

2	1
---	---

3	1
---	---

4	2
---	---

5	3
6	5
7	8
8	13

Each number is the **sum of the previous two numbers**.

 Formula:

$$F(n) = F(n-1) + F(n-2)$$

Example

$$2 = 1 + 1$$

$$3 = 2 + 1$$

$$5 = 3 + 2$$